

Positional Semantics for Retrieved Native-KV Memory: RoPE, Fragmentation, and Bounded Relocation in Progressive Retrieval Attention

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Abstract

Retrieved native-key/value (K/V) memory reuses attention states encoded at another time or place. Which positions should cached keys represent when a bounded model encodes blocks independently and later attends to selected states at a different logical distance? We add a minimal position strategy to Progressive Retrieval Attention (PRA), preserving learned absolute positions as the default and supporting rotary position embedding (RoPE) in ordinary and retrieved-memory attention. Five-seed controlled tiny and small models learn a fixed-target synthetic task to 100% held-out answer-token accuracy before parameter-preserving conversion to native-KV PRA. RoPE common-translation invariance holds numerically through an 8,192-token shift (maximum attention-logit RMSE 5.8×10^{-5}). Logical global positions eliminate layer-0 K fragmentation for both mechanisms; deeper error remains contextual and falls with overlap. At 64 source splits, oracle K/V recovers 0.949 of dense-context benefit for tiny RoPE, versus 0.483 for tiny absolute positions, but this direction reaches only the minimum five-seed two-sided sign result, $p = 0.0625$. Small-model oracle results are strong for both mechanisms. Routing and all-memory conditions remain unstable, and bounded-head means are influenced by seed outliers. RoPE makes native-KV relocation mathematically clean, but does not remove contextualization or selection failure. The results provide a reproducible basis for pretrained-model integration rather than evidence of production long-context quality.

1 Introduction

A transformer key is not only a content representation. Depending on the architecture, it also carries or interacts with position. This is easy to ignore while all keys are produced in one dense forward pass. It becomes operational when a system encodes a long source in bounded blocks, stores native per-layer K/V, retrieves a sparse subset, and combines it with a later query. The source key may have a local block position, a logical document position, or a position assigned at materialization time. These choices are not equivalent.

PRA provides a controlled setting for studying the problem. It stores full layer-native K/V behind inexpensive routing gists and inserts selected states before direct K/V under one softmax. Paper 1 established native transport, exact routing, model-bounded encoding, and an implicit prompt-head reference. The present paper holds those mechanisms fixed and changes the positional semantics.

Our contributions are:

- a reusable position strategy supporting learned absolute, sinusoidal, and RoPE semantics without changing the default absolute checkpoint layout;

- a weight-preserving RoPE SelfAttention-to-PRA conversion whose no-memory logits match the source model;
- explicit position metadata and post-position state labels on cached native K/V;
- probes separating pure translation, positional reset, contextual block fragmentation, overlap, pre-versus post-RoPE K, and bounded implicit-head behavior; and
- five-seed tiny/small evidence, including negative routing and variance results that constrain the claims.

2 Background and Related Work

Learned absolute embeddings add a position-dependent vector before the decoder stack and therefore entangle content and position in later hidden states. Transformer-XL introduced relative positional treatment for recurrent segment memory [2]. Shortformer moved absolute position information to queries and keys to improve cached-state reuse across subsequences [4]. ALiBi adds distance-dependent attention biases for train-short, test-long extrapolation [5].

RoPE rotates query and key feature pairs by position-dependent phases, making their dot product depend on relative displacement while each vector retains an absolute phase [6]. Position Interpolation rescales indices into a trained range [1]; LongRoPE uses nonuniform interpolation and progressive extension [3]. Those methods enlarge dense model context. Our question is different: how should positions behave when only selected externally stored native K/V is returned? StreamingLLM preserves selected chronological cache states and attention sinks [7]; PRA retrieves URI- or request-addressed chunks, so it must also specify source positions, relocation, and routing semantics.

3 Positional Semantics for Native K/V

3.1 RoPE and common translation

For one head of width d_h , let R_p be the block-diagonal rotation at position p :

$$\tilde{q}_i = R_i q_i, \quad \tilde{k}_j = R_j k_j. \quad (1)$$

Orthogonality and the group property give

$$\tilde{q}_i^\top \tilde{k}_j = q_i^\top R_i^\top R_j k_j = q_i^\top R_{j-i} k_j. \quad (2)$$

Thus translating both positions by c leaves the score unchanged: $(R_{i+c} q_i)^\top (R_{j+c} k_j) = q_i^\top R_{j-i} k_j$. This supports relocation of a complete query-memory relation. It does not imply that moving only a cached key leaves attention unchanged; a fixed query should observe the changed relative distance.

3.2 Three independent choices

Three choices are often collapsed into "RoPE support":

1. **Encoding positions:** local block indices or logical source indices.
2. **Storage state:** pre-positional K_{raw} plus metadata, or published post-RoPE K_{rope} .

3. **Materialization policy:** preserve source distance, reset locally, relocate as a block, compress distance, or retain order only.

The canonical implementation publishes post-position K. Pre-position storage remains experimental because deferred rotation changes the cache representation and runtime work.

3.3 Position and contextualization fragmentation

If a dense source is split into bounded blocks, local position reset changes absolute embeddings and post-RoPE key phases. Independently, later blocks lose causal left context, which changes hidden states even with perfect position IDs. We call these *positional fragmentation* and *contextualization fragmentation*. Logical global positions can remove the first at layer 0. Encoding overlap can only approximate the second.

4 Implementation

The shared code adds a registry-backed `PositionEncoding` interface. Its first method positions token embeddings; its second transforms per-head Q/K with shape $[B, H, T, d_h]$. Absolute mode adds the existing learned table and leaves projected Q/K unchanged. RoPE mode leaves token embeddings unchanged and rotates Q/K. A position-aware SelfAttention block is instantiated only for non-absolute modes, so old absolute models retain their original `TransformerEncoderLayer` state names and behavior.

PRA attention applies the same strategy in direct and cache-publication paths:

```
q = split_heads(q_proj(hidden))      # [B,H,T,Dh]
k = split_heads(k_proj(hidden))      # [B,H,T,Dh]
q, k = position.transform_qk(q, k, position_ids)

k_ref = split_heads(k_proj(reference_hidden))
_, k_ref = position.transform_qk(k_ref, k_ref, reference_positions)
cache.publish(k_ref, v_ref, position_ids, state="post_position")
```

Values are not rotated. Each `LayerKV` records positions used to publish K and a `position_state`. Slicing encoding blocks into routing chunks preserves this metadata, as does CPU residency. RoPE IDs may exceed the model’s native token count; the token count in any one base-model operation remains bounded.

Conversion copies Q/K/V projections, output projection, norms, feed-forward layers, embeddings, and language-model head from trained RoPE SelfAttention into PRA. With memory disabled, source and converted logits match to unit-test tolerance. No memory adapter or additional transport projection is introduced.

For a long prompt, the adapter keeps a direct tail and registers displaced tokens under `#_head`. Bounded encoding produces native K/V, smaller chunks produce routing gists, and whole chunks are selected under

$$T_{\text{direct}} + T_{\text{memory}} + T_{\text{reserve}} \leq T_{\text{native}}. \quad (3)$$

RoPE continues logical IDs past T_{native} without enlarging an operation.

5 Experimental Design

5.1 Models and training

Controlled tiers are smaller than repository deployment profiles so all paired seeds fit one NVIDIA GTX 950M. Both modes retain the learned position table, unused by RoPE, to match parameter count exactly.

Table 1: Controlled matched model tiers.

Tier	Layers	Heads	d	d_{ff}	Parameters	Steps
Tiny	2	4	64	128	92,288	100
Small	4	4	128	256	580,608	150

Seeds are $\{1, 7, 21, 42, 87\}$. Each absolute/RoPE pair uses the same 2,048 fixed-target synthetic training examples, optimizer, batches, vocabulary, 192-token native context, and training tokens. Every run reaches 100% held-out answer-token accuracy. The downstream set has 12 invariant source/query/answer examples; only source boundaries change. We evaluate 2, 5, 16, 32, and 64 splits.

PRA uses post-position K, local positions per independent reference, a 160-token whole-chunk budget, and a four-token reserve in the split experiment. Conditions include dense full context, tail only, native all, evidence oracle, gist-routed, shuffled, and memory-disabled. Recovered context benefit is

$$\text{RCB} = \frac{\mathcal{L}_{\text{tail}} - \mathcal{L}_{\text{PRA}}}{\mathcal{L}_{\text{tail}} - \mathcal{L}_{\text{full}}}, \quad (4)$$

reported without clipping.

5.2 Additional probes

The pure probe fixes Q/K/V across shifts up to 8,192 and measures logit RMSE, Jensen–Shannon divergence, top-token agreement, and output RMSE. The pre/post probe compares stored post-RoPE K with deferred rotation at identical positions, K-only relocation against a fixed query, and common Q/K translation.

The decomposition compares dense 128-token encoding with 32-token blocks under local reset, logical global IDs, and 5–50% left overlap. The bounded-head probe inserts neutral tokens between source evidence and query, evaluates 1L, 2L, 4L, and 8L logical lengths, keeps 32 tokens direct, permits 160 memory tokens, and records the largest native call.

6 Results

6.1 Pure translation and storage

Across head dimensions 8–64, common translation preserves the top-attended token exactly. Maximum logit RMSE is 5.79×10^{-5} , maximum JS divergence is 3.00×10^{-7} , and maximum output RMSE is 1.80×10^{-5} (Figure 1).

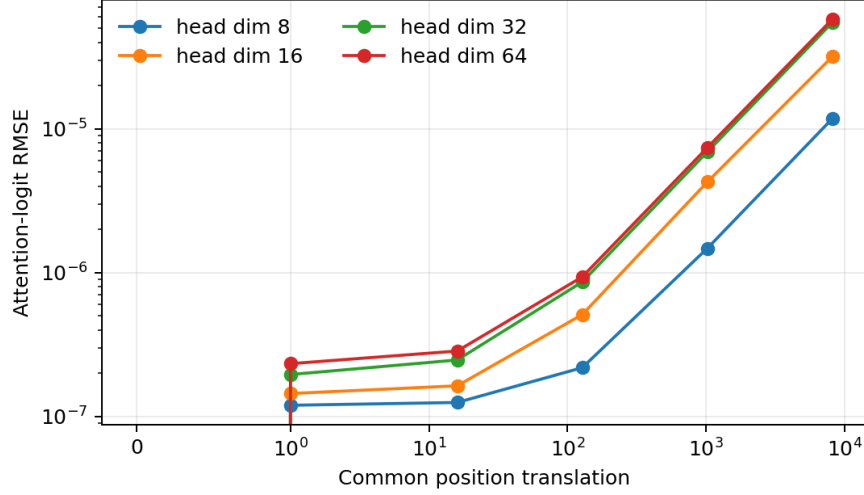


Figure 1: Numerical RoPE common-translation invariance. The logarithmic vertical axis exposes floating-point error, not a quality effect.

Publishing post-RoPE K and rotating raw K later at the same positions matches exactly. Moving K alone by 128–8,192 positions changes fixed-query scores (mean RMSE 7.14–8.11), as relative distance should. Translating Q and K together remains near invariant (maximum RMSE 4.91×10^{-4} at 8,192). Int64 position metadata costs 0.91% of fp32 K/V bytes in these shapes. Pre-position storage is plausible, but this does not establish its end-to-end latency or quality advantage.

6.2 Split scaling

Figure 2 reports the trained experiment. Split 2 often creates a chunk too large for the 160-token whole-chunk budget, so native-all becomes tail-only. This is a materialization outcome, not a positional failure.

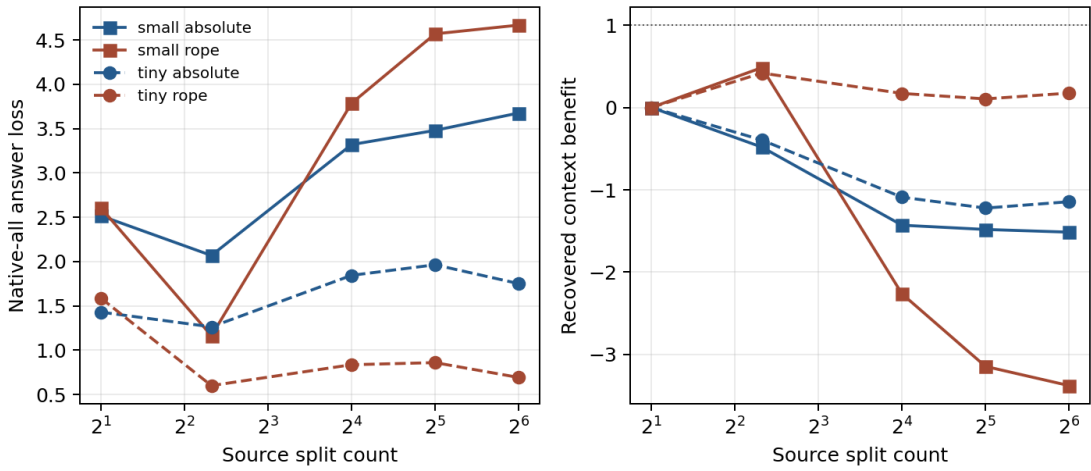


Figure 2: Five-seed native-all loss and recovered context benefit. Negative RCB means that unfiltered memory hurts relative to tail-only inference.

Table 2: Mean native-all loss by split count. Lower is better.

Tier	Position	5	16	32	64
Tiny	Absolute	1.264	1.845	1.964	1.752
Tiny	RoPE	0.601	0.837	0.860	0.693
Small	Absolute	2.069	3.321	3.481	3.677
Small	RoPE	1.163	3.787	4.570	4.669

Tiny RoPE is lower in all five paired seeds at 16 and 32 splits. With five seeds, however, the smallest exact two-sided sign probability is 0.0625. At 64 splits it is lower in four of five. The small tier reverses in mean at 16–64 and has high seed variance. There is no scale-independent RoPE advantage.

Oracle selection isolates transport and sparsity from routing. At 64 splits, tiny absolute/RoPE oracle losses are 0.581/0.070, corresponding to RCB 0.483/0.949; RoPE is lower in all five seeds. Small oracle losses are both low, 0.126/0.046, with RCB 0.967/0.929. Routed losses remain much larger (tiny 1.994/1.301; small 4.434/4.721). Strong oracle outcomes beside weak routed outcomes locate much of the remaining failure in selection, not transport.

6.3 Position reset, context loss, and overlap

Logical global IDs make layer-0 K exactly match dense encoding for both mechanisms. Averaged over layers and seeds, local-reset versus logical-global K RMSE falls from 0.671 to 0.070 for tiny absolute and 0.653 to 0.062 for tiny RoPE. Small reductions are 0.683 to 0.166 and 0.625 to 0.171. The deeper-layer remainder is contextualization loss.

Fifty-percent overlap processes 1.375 times as many tokens and reduces mean K RMSE to 0.046/0.031 for tiny absolute/RoPE and 0.119/0.107 for small absolute/RoPE. The trend is not perfectly monotonic in every deep layer, so overlap is a cost-quality control, not a repair.

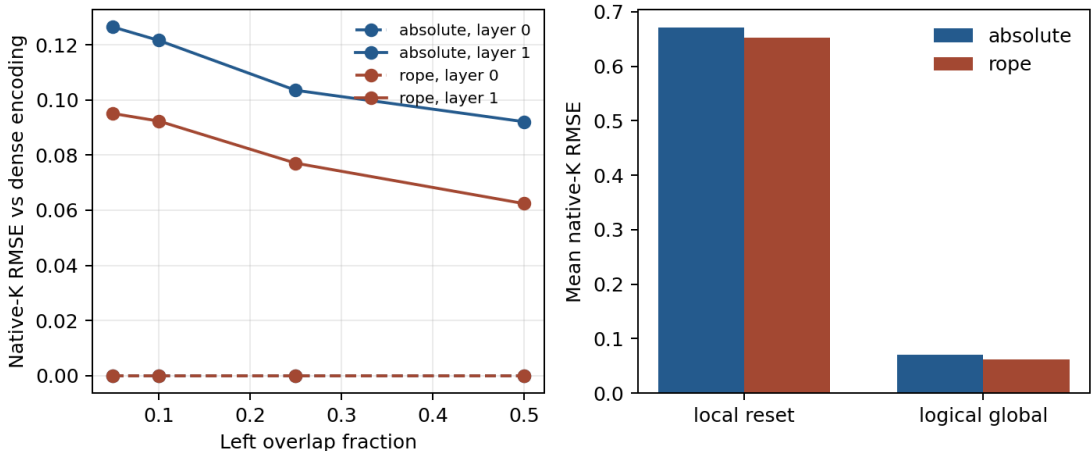


Figure 3: Dense-K deviation under position reset and overlap. Layer 0 becomes exact with logical positions; deeper layers retain context-boundary error.

6.4 Bounded implicit-head scaling

All 1L–8L runs obey the 192-token native-operation limit; violations are zero. The largest call is 160 tokens at 1L and 192 thereafter. Table 3 gives mean and median over five seed means, each computed from four examples.

Table 3: Implicit-head answer loss. Each cell is mean / median over five seed means.

Tier	Position	1L	2L	4L	8L
Tiny	Absolute	.143/.104	.805/.920	1.338/1.382	1.206/1.403
Tiny	RoPE	1.043/.734	.543/.017	.830/.032	1.023/.565
Small	Absolute	.002/.002	1.520/.002	1.524/.004	1.618/.005
Small	RoPE	.001/.001	.062/.013	.129/.011	.282/.042

Small absolute means are dominated by one catastrophic seed; four of five seeds remain near zero. RoPE avoids that catastrophe but is lower in only two of five paired seeds at 2L–8L. Tiny results also change direction by seed and distance. This is reduced catastrophic variance in this sample, not a robust quality win. RoPE memory benefit over its own disabled condition is positive at 2L and 4L but negative at 8L: positional stability can coexist with unhelpful retrieval.

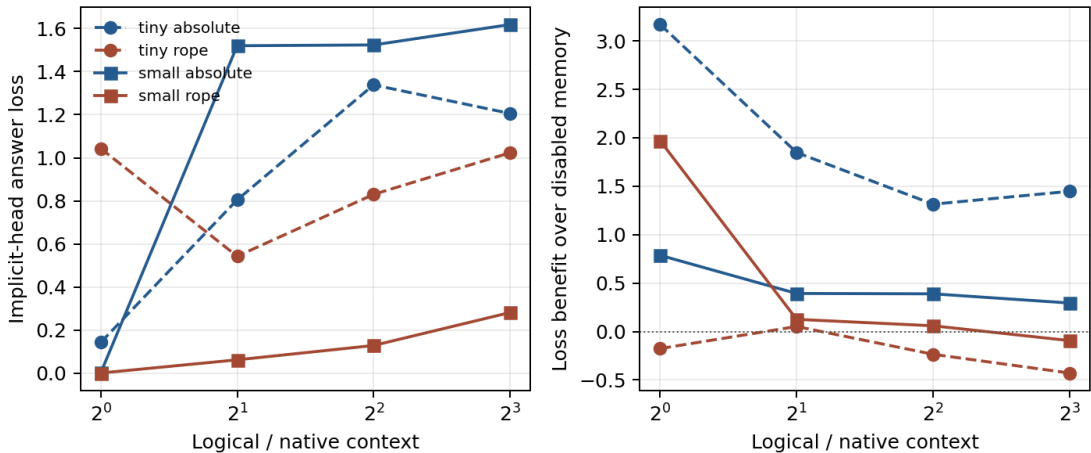


Figure 4: Bounded-head quality and benefit over disabled memory. Mean curves must be read with Table 3 because seed outliers are large.

7 Discussion

What RoPE solves. It applies positions directly to Q/K, admits logical indices beyond a native token limit, and provides exact common-translation semantics in real arithmetic. Raw K plus positions can be rotated later without changing same-position results.

What RoPE does not solve. It does not restore left context lost by independent encoding, choose the relevant chunk, prevent harmful attention to irrelevant K/V, or guarantee extrapolation from short training contexts. The small native-all reversal and weak routed conditions are counterexamples to a blanket "RoPE fixes PRA" claim.

Post- versus pre-position K. Post-position K is immediately compatible with attention but binds the cache to source positions. Pre-position K is portable and metadata is small, but deferred rotation adds materialization work and requires a distance policy. A future systems experiment must measure synchronized warm latency, cache compatibility, and accuracy under fixed, compressed, bucketed, and order-only policies.

Engineering implication. Position IDs belong in cache provenance even when K is post-position. Without them, a consumer cannot audit local versus logical coordinates, reproduce a score, or safely introduce deferred rotation.

8 Limitations and Threats to Validity

These are educational controlled models, not pretrained LLMs. The character tokenizer, binary task, 12 split examples, four head examples, and GTX 950M limit external validity. Five seeds permit paired direction checks but weak exact significance; an all-same-direction effect still has two-sided $p = 0.0625$. Mean loss is sensitive to rare catastrophes. The split benchmark uses independent references with effective local positions; logical-global results come from the decomposition. Pre-position storage is not the canonical runtime path. We do not report production throughput, large-model memory savings, natural-language QA, or unrestricted streaming.

9 Reproducibility

The pre-change baseline is tagged `paper1-pre-rope-baseline`; development occurs on `research/paper1-5-rope`. The measured code checkpoint is `2b50f34cbf9f42f5dec596f26493f8d67cfaf43d`. Raw JSON/CSV, plots, environment metadata, training histories, and manifests live under `docs/papers/shared/results/paper1.5_rope/`. Checkpoints are under `out/paper1.5_rope/`. Tests cover translation, norm preservation, large logical offsets, absolute bounds, cache metadata, and exact no-memory SA-to-PRA conversion. Existing Paper 1 tests remain the absolute-mode guardrail.

10 Conclusion

Retrieved native-KV memory needs an explicit positional contract. RoPE supplies clean relative geometry and portable logical indices, but model-bounded retrieval still has separate positional, contextual, and selection problems. Logical positions remove first-layer positional fragmentation; overlap reduces deeper contextual fragmentation; oracle K/V often restores dense benefit; and routing remains limiting. Pretrained integration is justified only with larger tasks, stronger routing, and pre/post-K runtime measurements.

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